ALICE L WALTON SCHOOL OF MEDICINE

Autonomic Nervous System – Vasopressors and Inotropes

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Learning Objectives (5)

After completing this brick, you will be able to:

1. Describe the functional classifications of the Autonomic Nervous System and their roles.
2. Classify autonomic receptors based on their G protein class and explain the associated G-protein-linked second messenger pathways.
3. Describe the rgic receptors and cholinergic receptors.
4. Predict the organ-specific effects resulting from activation and inhibition of adrenergic and cholinergic receptors based on the tissue expression patterns.
5. Explain the select systemic effects of specific vasopressors and inotropes.

CASE CONNECTION

You were performing a required interprofessional activity with an EMS group. They received a call for a 65-year-old male named TY who was found lying in his bathroom by his wife. The wife immediately dialed 911. Your "ride-along" EMS ambulance was dispatched. Upon arrival, EMS noted that TY was barely breathing, and his pulse was barely detectable. The EMS crew immediately started high-quality CPR, but no rhythm or HR was detected. They decided to give TY epinephrine and immediately injected the medication. They kept performing CPR, but no rhythm was restored. They gave another dose of epinephrine and continued CPR. EMS then detected a heart rhythm and his pressure of 50/30 mm Hg.

What was the pharmacological rationale behind the administration of Epinephrine to TY? Consider the answer to this question as you read the brick.

GO TO CONCLUSION ↓

What is the functional role of the Autonomic Nervous System? (LO#1)

We have discussed the Autonomic Nervous System (ANS) in the Autonomic Nervous System brick. In the upcoming weeks, we are going to integrate the ANS physiology to pharmacological actions mediated by the activation or inhibition of the ANS. Feel free to refer back to the physiology brick for more detailed information. First, let's begin with a quick recap of the ANS.

Based on the functional divisions of the nervous system, the peripheral nervous system is divided into the autonomic and somatic nervous system. The ANS is further divided into parasympathetic nervous system (PSNS), sympathetic nervous system (SNS), and enteric nervous system (ENS).

You can recall from Dr. Murray's brick that the SNS is the "fight or flight response" while the PSNS is the rest and digest stimulus. SNS and PSNS are 'pulling' each other and often have opposing functions (Figure 2). This is important to remember when we discuss the actions on the effector organs.

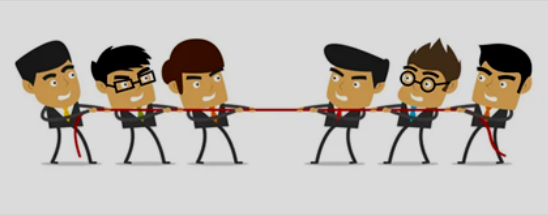


Figure 2: SNS and PSNS opposing functions

What are the functions of ANS?

Sympathetic Nervous System (SNS)

The changes that the the "flight or fight" response. It is the body's way of reacting to emergency situations. We know that when these situations occur, the adrenal medulla releases epinephrine (adrenaline) into the bloodstream and promotes the responses at the effector organs. The SNS functions as a 'complete system', as in a diffused distribution through postganglionic fibers. For example, when we exercise, the heart rate increases, blood pressure increases, and muscle metabolism increases simultaneously. The SNS also prepares the body for any unknown situation that might require it for survival. For instance, reaction to an unexpected encounter with a bear during a pleasant hike.

Parasympathetic nervous system (PSNS)

The PSNS maintains homeostasis within the body; in other words, it performs the functions that are essential for life. For instance, it initiates digestive processes and elimination of waste through the GI or Kidneys. The key distinction here is that it does not discharge as a unit. For example, think about how unnerving and unpleasant it would be to have a urination and defecation response at the same time. Thus, in contrast to SNS, the PSNS innervations are more direct and affect specific organs.

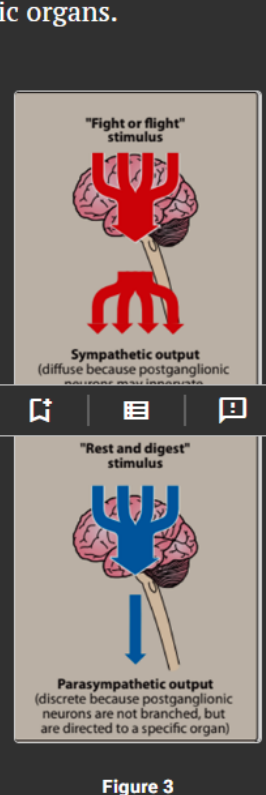

"Fight or flight" stimulus
Sympathetic output (diffuse because postganglionic neurons may innervate many organs)
"Rest and digest" stimulus
Parasympathetic output (discrete because postganglionic neurons are not branched, but are directed to a specific organ)

Figure 3

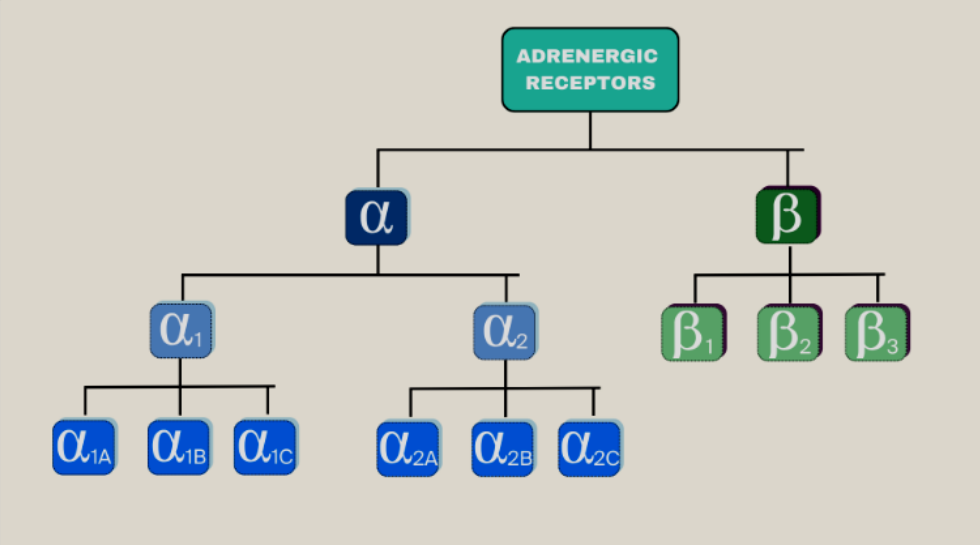
How are the neurons communicating with each other and effector organs?

Communication between the neurons and/or with the effector organs occurs through neurotransmitter release. Recall that the activation of an action potential leads to depolarization of the neurons and the release of the neurotransmitter. The type of neurotransmitter released at the synapse depends on the effector organ and the stimulus. The two main neurotransmitters that we will discuss are:

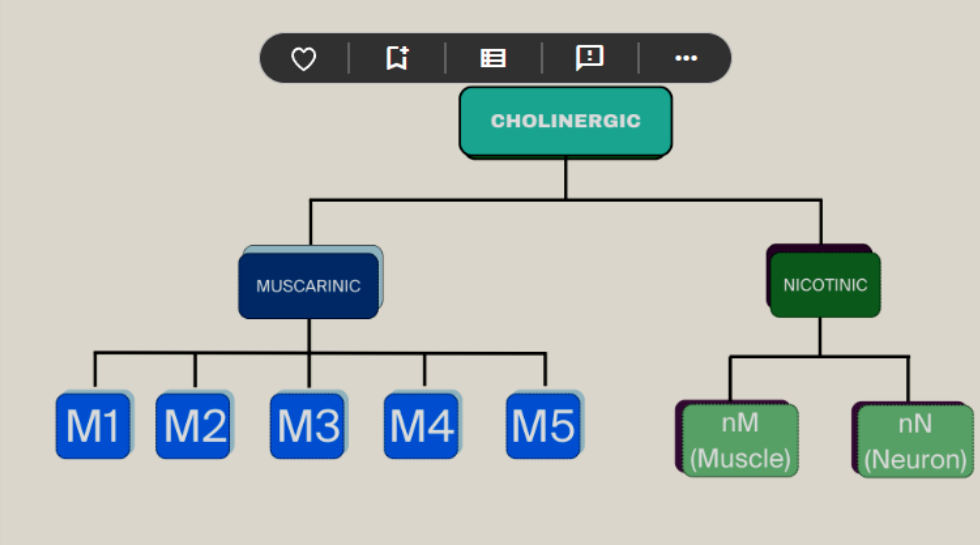
- **Acetylcholine (ACh):** The release of ACh occurs at the autonomic ganglia in both the SNS and PSNS. ACh is the primary neurotransmitter released in the autonomic postganglionic nerves that innervate effector organs. Hence, if ACh mediates the transmission, the neuron is termed as **cholinergic.**
- **Norepinephrine and epinephrine:** When norepinephrine or epinephrine is involved as the neurotransmitter, the nerve fiber is referred to as **adrenergic**. In the SNS, norepinephrine mediates most neurotransmission to the effector or adrenal medulla, is released directly into the bloodstream to target the effector organ.

How do we classify ANS receptors?

ANS receptors are classified into adrenergic and cholinergic receptors. Each of these are further classified into different receptor isoforms. The figure below describes ANS receptor classification.



Classification of Adrenergic Receptors



Classification of Cholinergic receptors

The neurotransmitters that propagate the signaling cascades, such as epinephrine and ACh are hydrophilic. They have a difficult time crossing the lipid bilayer. The neurotransmitters thus relay the stimulus to the effector organ to elicit a response by activating cell surface receptors. Thus, the neurotransmitter can be a signal that activates a signal detector (or receptor) and transducer. Other cellular proteins or molecules, known as second messengers, further propagate the response. (If this sounds familiar, recall from Signal transduction brick – BOOM).

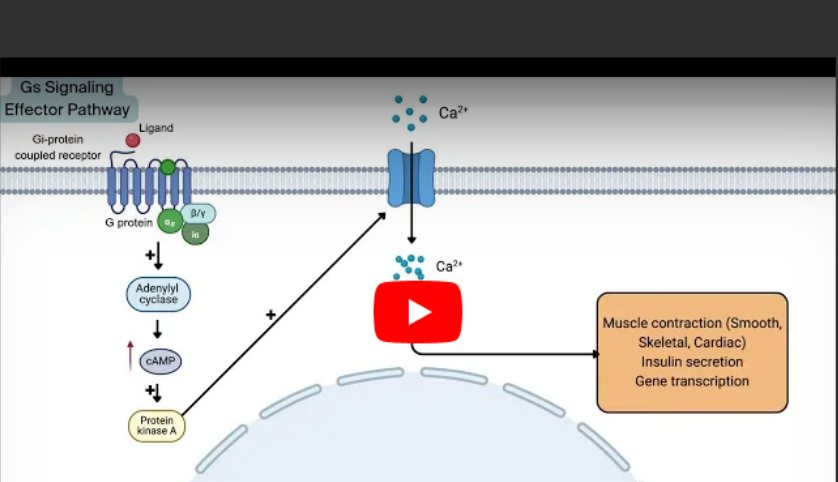
How are GPCRs classified? (LO#2)

All adrenergic and cholinergic receptors are isoforms of the various G-protein coupled receptors (Gi ways results in recruiting 2nd messengers that help generate a physiological response. For instance, Ca^{2+} is the 2^{nd} messenger (cellular activation) that is responsible for cardiac muscle contraction (a physiological response).

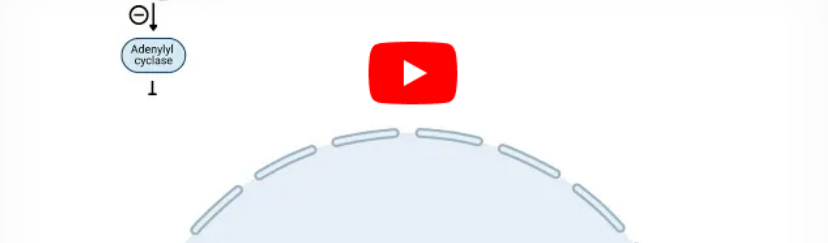
First, let's look at the various GPCRs and their signaling, then we can assign each subtype to the adrenergic and cholinergic receptors.

There are three types of GPCRs. These are **Gs, Gi, and Gq**. They all activate different 2^{nd} messengers to propagate signaling.

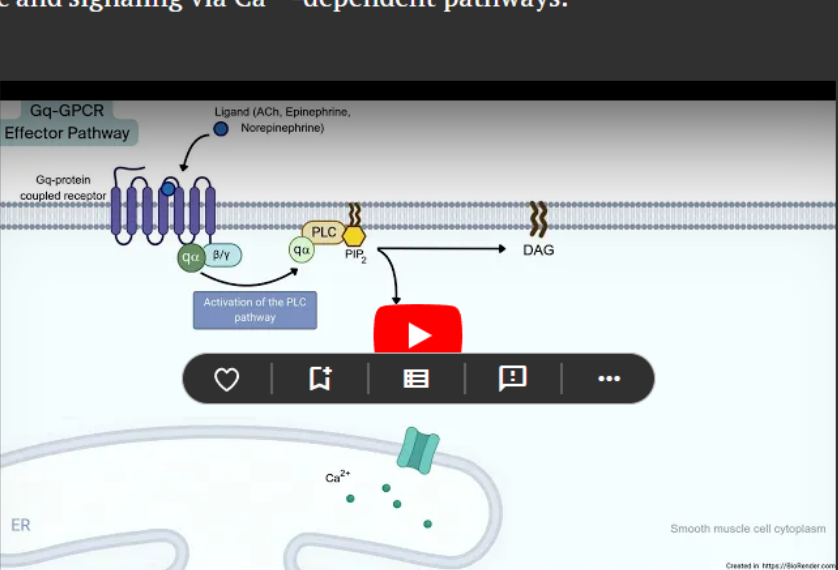
First, let's look at messaging that occurs with the Gs or G-stimulatory pathway. The messenger molecule, cAMP, get activated and carries out cellular response.


Gs Signaling Effector Pathway
Ligand
G-protein coupled receptor
G-protein
Adenylyl cyclase
cAMP
Protein kinase A
Ca^{2+}
Muscle contraction (Smooth, Skeletal, Cardiac)
Insulin secretion
Gene transcription

The Gi pathway (G-i hway. In this case, the messenger molecule, cAMP production in inhibited. (Opposite of Gs)

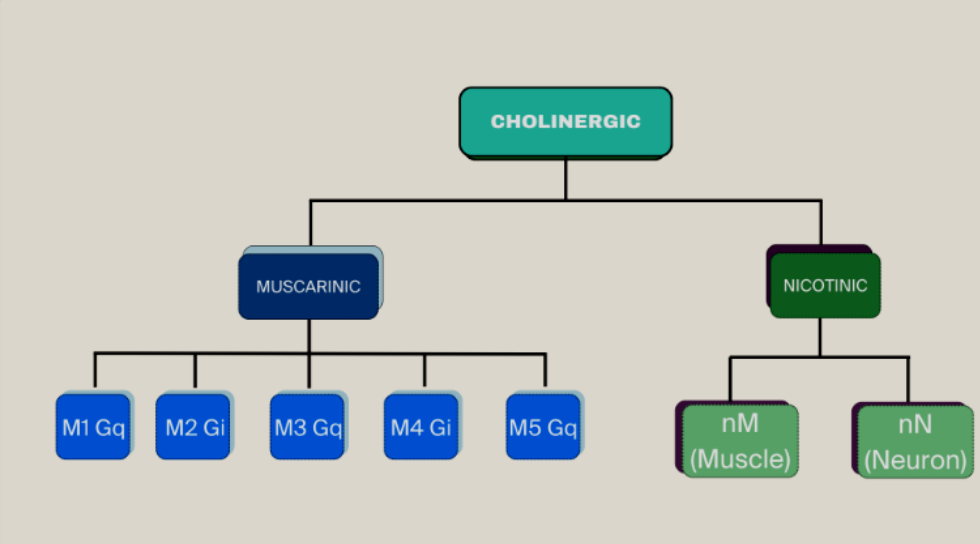


The Gq pathway differs from Gi and Gs. It involves activation of the Ca^{2+} release and signaling via Ca^{2+}-dependent pathways.


Gq-GPCR Effector Pathway
Ligand (ACh, Epinephrine, Norepinephrine)
Gq-protein coupled receptor
PLC
PIP2
DAG
Activation of the PLC pathway
Ca^{2+}
ER
Smooth muscle cell cytoplasm

How are the GPCRs related to adrenergic and cholinergic receptors?

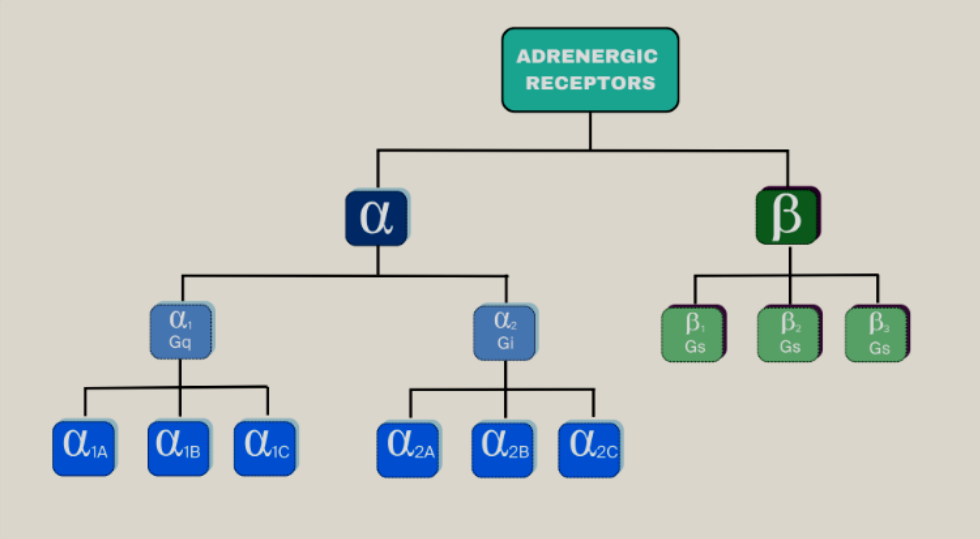
The figure below shows the adrenergic and cholinergic receptors and the coupled G-protein that regulate the effector proteins.



Cholinergic receptors and the associated GPCR

Key highlights

- There is no Muscarinic receptor that is a Gs-type of GPCR
- M2 and M3 receptor GPCR are important to remember (works in the heart and lungs!!)
- Nicotinic receptors are ligand-gated channels activated by ACh



Adrenergic Receptors and the associated GPCR

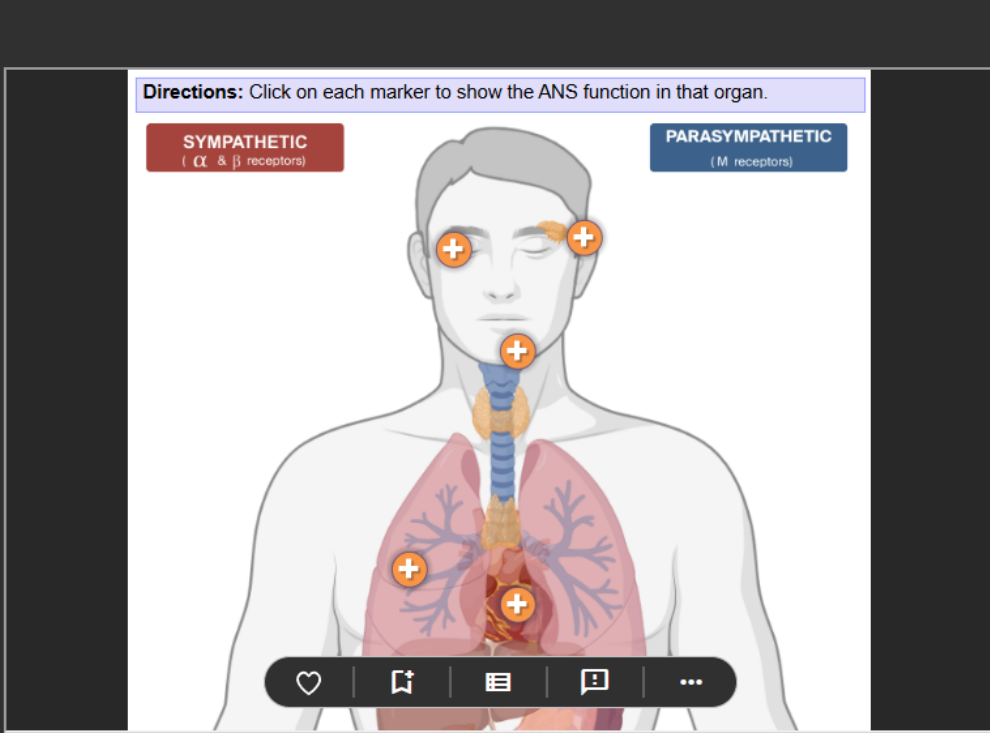
Key highlights

- All α1 receptor subtypes are Gq-type GPCR
- All β receptors a

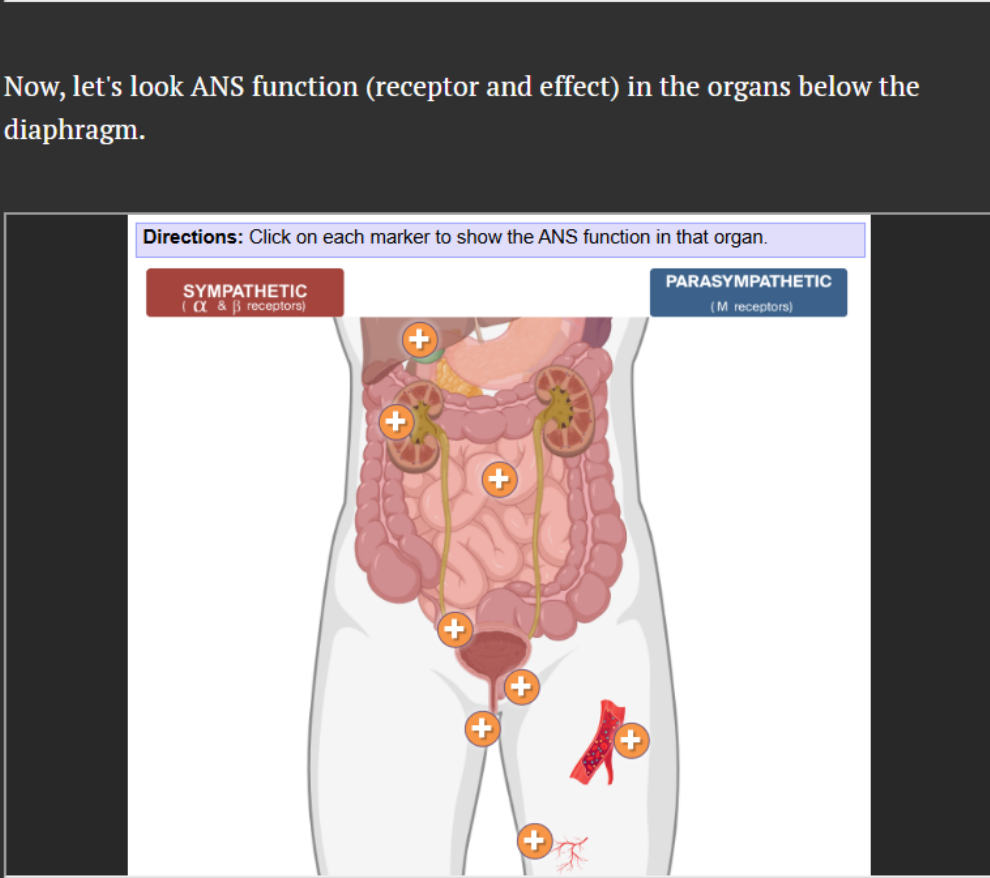
What happens when these receptors are activated? (LO#3 and 4)

To perform all our bodily functions, all organs express these autonomic receptors. Activation of these receptors elicits the physiological response. Remember the constant tug-of-war between the adrenergic and cholinergic systems! In most cases, the effect of adrenergic and cholinergic activation is the opposite. Let's *break it down* by looking the organs from head to the thorax.

First, let's look ANS function (receptor and effect) in the organs above the diaphragm.


Directions: Click on each marker to show the ANS function in that organ.
SYMPATHETIC (α & β receptors)
PARASYMPATHETIC (M receptors)

Now, let's look ANS function (receptor and effect) in the organs below the diaphragm.


Directions: Click on each marker to show the ANS function in that organ.
SYMPATHETIC (α & β receptors)
PARASYMPATHETIC (M receptors)

What drugs/agents can activate autonomic receptors? (LO#5)

In the previous sections, we have discussed the actions of endogenous neurotransmitters tha thetic discharge. Synthesized forms of vatives can be used as pharmacological agents to achieve a similar physiological effect. These synthesized neurotransmitters (adrenergic and cholinergic) or their derivatives can be classified as *autonomic agents (or dugs)*. Since these agents achieve the same effect as the endogenous ligands, they are also known as **adrenergic or cholinergic agonists.**

In this brick, we will introduce some of the adrenergic agonists that are commonly used in practice. Agonists that activate the adrenergic receptors are often used as **adrenergic agonists.** Since they activate responses that mimic the sympathetic nervous system (SNS), they are also called **sympathomimetics** (sympatho = SNS, mimetics= mimicking the sympathetic system).

The next section discusses some of the commonly used sympathomimetics agents and their hemodynamic effects given in therapeutic doses.

Epinephrine

When given at a therapeutic dose, Epinephrine interacts with both α and β receptors. It has **equal affinity for α and β receptors.** It increases cardiac contractility and heart rate by activating β1 receptors. Since, it has positive inotropic and chronotropic effect, it increases stroke volume and cardiac output). It increases systolic blood pressure by activation of α1 receptors and vasoconstriction of vascular beds. Epinephrine is helpful in bronchodilation (β2 activation) in the lungs. In the skeletal muscle, epinephrine induces vasodilation, hence diastolic blood pressure might fall.

CLINICAL REASON

Epinephrine is a cornerstone medication in Advanced Cardiac Life Support (ACLS) protocols, primarily used during cardiac arrest to improve the chances of return of spontaneous circulation (ROSC). It works by stimulating alpha and beta-adrenergic receptors, which leads to increased heart rate, enhanced myocardial contractility, and vasoconstriction. This vasoconstriction helps to improve blood flow to vital organs, particularly the heart and brain, during cardiopulmonary resuscitation (CPR). Administered in 1 mg doses intravenously every 3-5 minutes during resuscitation efforts, epinephrine is crucial for managing arrhythmias such as asystole, pulseless electrical activity (PEA), and other life-threatening arrhythmias. Its timely administration can significantly impact patient outcomes in emergency situations.

Norepinephrine

When given at therapeutic dose, Norepinephrine has more **α1 effects than α2** or the β receptors. Hence it has more affinity for **α1>>α2**. This causes constriction of all vascular beds. This allows it to increase blood pressure (systolic and diastolic) and the systemic vascular resistance. It is non-selective for the β receptor activation.

What is the pri ood pressure?

Dopamine

When given in therapeutic doses, dopamine **activates β1** and increases cardiac contractility. Higher doses can also increase α1 effects. Dopamine also **activates D1 and D2** Dopaminergic receptors, which are in the renal vascular beds and produce vasodilation.

What receptors does dopamine activate in the renal vascular beds?

Dobutamine

When administered at therapeutic doses, dobutamine **activates β1 receptors.** Dobutamine has high affinity for the β1 receptors than β2 receptors. This β1-effect helps it increase cardiac contractility (positive inotropic) and heart rate (positive chronotropic). Heart rate, stroke volume, and cardiac output increase due to this β-1 mediated effect.

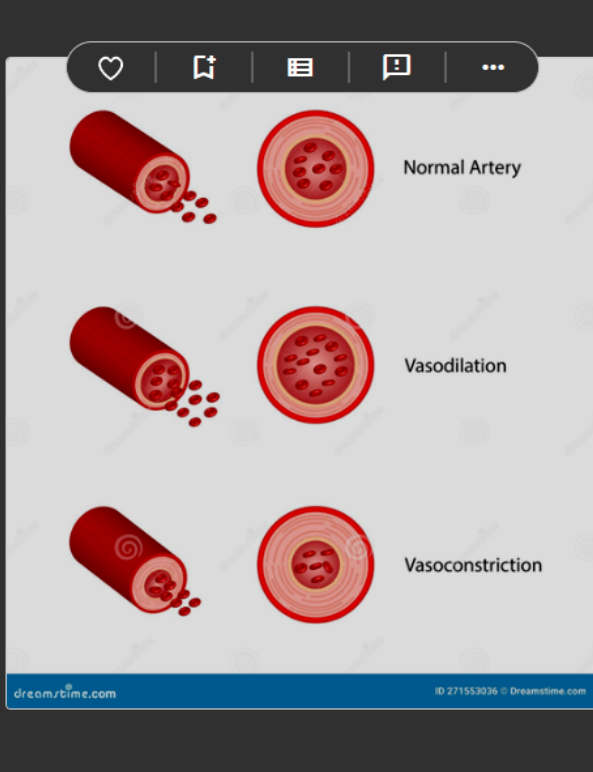
Isoproterenol

Isoproterenol is a potent activator of both **β1 and β2 adrenergic receptors** (β1=β2). It can increase heart rate and cardiac contractility. Cardiac output and heart rate are increase; however, these effects are less compared to dobutamine. Isoproterenol can also cause bronchodilation by activation of β2 receptors (relaxation of bronchial smooth muscles).

What are vasopressors and inotropes?

Vasopressors are pharmacological agents used to treat acute conditions. They function by increasing the vasoconstriction of blood vessels. As the name suggests, vaso = vasculature, pressor = pressing, so these agents increase the constriction of blood vessels.

When constriction occurs, the blood exerts more pressure on the walls, and hence blood pressure increases. An increase in blood pressure leads to an increase in vascular resistance. If the agent is administered systematically, the vascular resistance in the vasculature increases, leading to an increase in systemic vascular resistance (SVR). These agents are used in acute, emergent situations where the patients are not hemodynamically stable.



Inotropes enhance cardiac contractility and improve cardiac output (CO). The ion (of inotropy) refers to Ca^{2+} here. When intracellular Ca^{2+} increases, it binds to calmodulin, which increases muscle contraction. When this process occurs in cardiac cells, it increases the contraction of cardiac muscle and the force of cardiac contractility. This increase in cardiac contractility is referred to as an inotropy. Inotropic agents that increase contractility are categorized as **positive inotropic agents.** Similar to vasopressors, these agents are used in acute situations where the patient's hemodynamic stability needs to be restored.

Agonists that activate the adrenergic receptors are often used as vasopressors and inotropes. Thus, o be classified into vasopressors or inotropes.

In practice, epinephrine, norepinephrine, and dopamine are typically classified as vasopressors. These agents have potent effects on the vasculature and hence are known as vasopressors. Dobutamine, on the other hand increases the contractility (or increases inotropy) and hence is classified as an inotrope or a positive inotropic agent.

CASE CONNECTION

BACK TO INTRODUCTION ↑

The pharmacological rationale for administering epinephrine in this situation is its ability to stimulate both alpha and beta-adrenergic receptors, which have several beneficial effects during cardiac arrest. Epinephrine is a critical medication used to manage asystole (no electrical and mechanical activity of the heart), as seen in TY's case. Firstly, epinephrine's a1-adrenergic receptor stimulation causes vasoconstriction, increasing systemic vascular resistance and the blood pressure. This vasoconstriction helps to improve blood flow to vital organs, particularly the heart and brain. Secondly, the stimulation of beta-adrenergic receptors by epinephrine increases heart rate and cardiac contractility. This can help to restart the heart and improve its pumping efficiency once circulation is restored. Additionally, beta-adrenergic stimulation can enhance the electrical activity of the heart, which is crucial in cases of asystole where there is no detectable electrical activity.

Summary

- The Autonomic N the sympathetic, parasympathetic, and enteric nervous systems, each with distinct functions.
- The sympathetic nervous system (SNS) is responsible for the 'fight or flight' response, involving widespread physiological changes like increased heart rate and blood pressure.
- The parasympathetic nervous system (PSNS) maintains homeostasis by regulating essential functions like digestion and waste elimination, with more targeted and specific effects.
- Neurotransmitters such as acetylcholine and norepinephrine mediate communication between neurons and effector organs, with receptors classified as adrenergic or cholinergic.
- Adrenergic agonists, or sympathomimetics, mimic SNS responses and are used as vasopressors and inotropes to manage acute hemodynamic instability.
- Vasopressors increase vascular resistance and blood pressure, while inotropes enhance cardiac contractility, both crucial in emergency medical situations.

Review Questions

1. Which of the following is an example of cholinergic activation

A. ● Contraction of the iris and pupils contract
B. ○ Increased contraction of the heart
C. ○ Dilation of the bronchial smooth muscles
D. ○ Decrease

Hide Explanation

The correct answer is Contraction of the iris and pupils contract (A). Cholinergic activation can cause pin-point pupils since the iris contracts. The increase in heart rate (B) is incorrect as cholinergic activation decreases heart rate by activation of M2 receptors. The dilation of bronchial smooth muscles (C) is incorrect since cholinergic activation increases bronchial contractions. The decrease in gastric secretions in stomach is incorrect since cholinergic activation increases acid production in the stomach. Cholinergic activations promotes the "rest and digest"function.

2. Which of the following describes the Gq-GPCR pathwway?

A. ○ Inhibition of adenylyl cyclase enzyme
B. ● Activation of IP3 and Ca2+ release
C. ○ Activation of cAMP by PLC-β
D. ○ Inhibition of cAMP by PKA

Hide Explanation

The correct answer is activation of IP3 and Ca2+ release (B). The activation of Gq-GPCR activates PLCβ s to its receptor and increases cytosolic Ca^{2+}. The inhibition of adenylyl cyclase (A) is incorrect, it occurs with the Gi pathway. The activation of cAMP by PLCβ (C) is incorrect; it is not a signaling pathway. The inhibition of cAMP by PKA (D) is incorrect, it occurs in the Gi GPCR pathway.

3. Which receptor does Norepinephrine have the most affinity within a therapeutic dose?

A. ● α1
B. ○ α2
C. ○ β1
D. ○ β2

Hide Explanation

The correct answer is α1 receptors (A). Norepinephrine primarily targets a1 receptors, leading to increased blood pressure and systemic vascular resistance. α2receptors (B) and β1 and 2 receptors (C, D) are less affected at therapeutic doses.

4. Which receptor does Dobutamine stimulate to cause inotropic effect?

A. ● β1
B. ○ β2
C. ○ α2
D. ○ β3

Hide Explanation

The correct answer is β1 (A). Dobutamine is a positive inotropic agent that has high affinity for the β1 receptors. The β1 effect increases cardiac contractility and hence helps in the inotropic effect. β2 (B) is incorrect since β2 is mostly in the lungs and helps with bronchodilation. α2 (C) is incorrect, dobutamine does not have much α2 mediated effect. β3 (D) is incorrect; they are mostly expressed in the urinary bladder and do not have any inotropic functions.

5. Which neurotransmitter is released by cholinergic neurons?

A. ● Acetylcholine
B. ○ Norepinephrine
C. ○ Epinephrine
D. ○ Dobutamine

Hide Explanation

The correct answer is Acetylcholine (A). Cholinergic neurons release acetylcholine (ACh) to mediate transmission. Norepinephrine (B) and epinephrine (C) are released by adrenergic neurons. Dobutamine is not produced endogenously and is not invovled in neurotransmission discussed here

6. Which receptor is responsible for increasing the heart rate?

A. ○ α1
B. ● β1
C. ○ β2
D. ○ M2

Hide Explanation

The correct answer is β1 receptors. Adrenergic β1 receptors are Gi-type GPCRs that increase cardiac muscle contraction and increase HR. Answer choice, α1 (A) is incorrect. α1-receptors are mostly expressed in the peripheral system and are involved in smooth muscle contraction. Adrenergic β2 receptors (C) are involved in smooth muscle relaxation and do not increase heart rate. Muscarinic receptor, M2 (D) is responsible for decreasing HR.

H k?

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the functional classifications of the Autonomic Nervous System and their roles.~~
02 ☑ ~~Classify autonomic receptors based on their G protein class and explain the associated G-protein-linked second messenger pathways.~~
03 ☑ ~~Describe the tissue expression pattern of the adrenergic receptors and cholinergic receptors.~~
04 ☑ ~~Predict the organ-specific effects resulting from activation and inhibition of adrenergic and cholinergic receptors based on the tissue expression patterns.~~
05 ☑ ~~Explain the select systemic effects of specific vasopressors and inotropes.~~

Objective Confidence: 5/5